

MATHEMATICS

TARGET : JEE- Advanced

CAPS-4**INVERSE TRIGONOMETRIC FUNCTION
& SOLUTION OF TRIANGLE****SCQ (Single Correct Type) :**

- The value of $\sin^{-1}(\cos 2) - \cos^{-1}(\sin 2) + \tan^{-1}(\cot 4) - \cot^{-1}(\tan 4) + \sec^{-1}(\operatorname{cosec} 6) - \operatorname{cosec}^{-1}(\sec 6)$ is
(A) 0 (B) 3π (C) $8 - 3\pi$ (D) $5\pi - 16$
- Let $\alpha = \cot^{-1}\left(\frac{\pi}{3}\right)$, $\beta = \sin^{-1}\left(\frac{\pi}{4}\right)$ and $\gamma = \sec^{-1}\left(\frac{2\pi}{3}\right)$, then the correct order sequence is
(A) $\alpha < \gamma < \beta$ (B) $\beta < \alpha < \gamma$ (C) $\gamma < \beta < \alpha$ (D) $\alpha < \beta < \gamma$
- Let $f(x) = \frac{2}{\pi} (\sin^{-1}[x] + \tan^{-1}[x] + \cot^{-1}[x])$ where $[x]$ denotes greatest integer less than or equal to x . If A and B denote the domain and range of $f(x)$ respectively, then the number of integers in $(A \cup B)$, is
(A) 1 (B) 2 (C) 3 (D) 4
- The radii of the escribed circles of $\triangle ABC$ are r_a , r_b and r_c respectively. If $r_a + r_b = 3R$ and $r_b + r_c = 2R$, then the smallest angle of the triangle is
(A) $\tan^{-1}(\sqrt{2} - 1)$ (B) $\frac{1}{2} \tan^{-1}(\sqrt{3})$ (C) $\frac{1}{2} \tan^{-1}(\sqrt{2} + 1)$ (D) $\tan^{-1}(2 - \sqrt{3})$
- If $\cos^{-1} \frac{x}{a} - \sin^{-1} \frac{y}{b} = \theta$ ($a, b \neq 0$), then the maximum value of $b^2x^2 + a^2y^2 + 2abxy \sin \theta$ equals
(A) ab (B) $(a + b)^2$ (C) $2(a + b)^2$ (D) a^2b^2
- In $\triangle ABC$, the bisector of the angle A meets the side BC at D and the circumscribed circle at E , then DE equals
(A) $\frac{a^2 \sec \frac{A}{2}}{2(b+c)}$ (B) $\frac{a^2 \sin \frac{A}{2}}{2(b+c)}$ (C) $\frac{a^2 \cos \frac{A}{2}}{2(b+c)}$ (D) $\frac{a^2 \operatorname{cosec} \frac{A}{2}}{2(b+c)}$
- If $[\cos^{-1} x] + [\cot^{-1} x] = 0$, where $[.]$ denotes the greatest integer function, then the complete set of values of x is
(A) $(\cos 1, 1]$ (B) $(\cos 1, \cot 1)$ (C) $(\cot 1, 1]$ (D) $[0, \cot 1)$

8. $f: \mathbb{R} \rightarrow (0, \pi/2]$ and $f(x) = \cot^{-1}(x^2 - 2ax + a + 1)$ is surjective than $a \in$
- (A) $\left\{ \frac{1-\sqrt{5}}{2}, \frac{1+\sqrt{5}}{2} \right\}$ (B) $\left\{ \frac{1-\sqrt{5}}{2}, \frac{1-\sqrt{5}}{2} \right\}$
- (C) $\left(-\infty, \frac{1-\sqrt{5}}{2} \right) \cup \left(\frac{1+\sqrt{5}}{2}, \infty \right)$ (D) None of these
9. In a $\triangle ABC$, if $b^2 + c^2 = 1999a^2$, then $\frac{\cot B + \cot C}{\cot A}$ is equal to ____.
- (A) $\frac{1}{999}$ (B) $\frac{1}{1999}$ (C) 999 (D) 1999
10. If $2a \cos B = c$ in a $\triangle ABC$ with sides a, b, c then $\tan \frac{A}{2} \left(\tan \frac{A}{2} + 2 \tan \frac{C}{2} \right)$ is equal to ____.
- (A) 1 (B) 21 (C) $\frac{1}{2}$ (D) $\frac{1}{3}$
11. In $\triangle ABC$, if $BC = 1$, $\sin \frac{A}{2} = x_1$, $\sin \frac{B}{2} = x_2$, $\cos \frac{A}{2} = x_3$, and $\cos \frac{B}{2} = x_4$ with
- $\left(\frac{x_1}{x_2} \right)^{2010} - \left(\frac{x_3}{x_4} \right)^{2009} = 0$, then length of AC is equal to ____.
- (A) > 1 (B) < 1 (C) $\frac{1}{2}$ (D) 1

MCQ (One or more than one correct) :

12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos^{-1}(\{x\})$ where $\{x\}$ is fractional part function. Then which of the following is/are correct?
- (A) f is many one but not even function. (B) Range of f contains two prime numbers.
- (C) f is aperiodic. (D) Graph of f does not lie below x -axis.
13. Which of the following statement(s) is/are **TRUE**?
- (A) Domain of $y = \cos^{-1}(e^x)$ is same as range of $y = -\sqrt{-x}$.
- (B) Number of elements common in the range of function $y = \tan^{-1}(\operatorname{sgn} x)$ and $y = \cot^{-1}(\operatorname{sgn} x)$ is only 1 (where $\operatorname{sgn} x$ denotes signum function of x .)
- (C) The function $y = \operatorname{sgn}(\cot^{-1} x)$ and $y = 1$ are identical functions.
- (D) Number of integers in the solution set of $1 < \log_2(\tan^{-1} x) < 2$ is 4.
14. Let a function $f: A \rightarrow B$ be defined as $f(x) = \sin^{-1}(\tan x) - \operatorname{cosec}^{-1}(\cot x)$. Which of the following statement(s) is/are **TRUE** for the function $f(x)$?
- (A) $f(x)$ is periodic with fundamental period π .
- (B) The function $f(x)$ is non-invertible.
- (C) The composite function $f(f(x))$ is not defined.
- (D) The function $f(x)$ is an even function.

15. Which of the following expression(s) have their value equal to four times the area of the triangle ABC? (All symbols used have their usual meaning in a triangle)
- (A) $rs + r_1(s - a) + r_2(s - b) + r_3(s - c)$ (B) $\frac{(a + b + c)^2}{\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2}}$
- (C) $(a^2 + b^2 - c^2) \tan B$ (D) $b^2 \sin 2C + c^2 \sin 2B$
16. Following the usual notations, in a triangle ABC, if $(\sqrt{3} - 1)a = 2b$ and $A = 3B$, then C cannot be equal to _____.
- (A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{6}$
17. Let ABC be a triangle with $\angle BAC = 120^\circ$ and $AB \cdot AC = 1$. Also, let AD be the length of the angle bisector of angle A of the triangle. Then _____.
- (A) Minimum value of AD is $\frac{1}{2}$ (B) Maximum value of AD is $\frac{1}{2}$
- (C) AD is minimum when $\triangle ABC$ is isosceles (D) AD is maximum when $\triangle ABC$ is isosceles

Comprehension Type Question:

Let α, β and γ are the roots of the equation $x^3 + 6x + 3 = 0$

and $A = \cos^{-1} \left(\sin \left((\alpha + \beta)^{-1} + (\beta + \gamma)^{-1} + (\gamma + \alpha)^{-1} \right) \right)$

$$B = \cos \left(\tan^{-1} \left(\sin \left(\frac{\alpha + \beta + \gamma}{2} \right) \right) \right)$$

$$C = \sec^{-1} \left(\operatorname{cosec} \left((1 - \alpha)(1 - \beta)(1 - \gamma) \right) \right).$$

18. If the range of the quadratic trinomial $g(x) = x^2 - 2Bx + k$ is $[0, \infty)$, then range of k equals
- (A) $[1, \infty)$ (B) $(1, \infty)$ (C) $\{1\}$ (D) $(-\infty, 1]$
19. The value of $(5A + B - C)$ is equal to
- (A) 1 (B) 10 (C) 5 (D) 0
20. Range of the function $f(x) = \frac{(5A - C)x^5 + 6Bx^2}{x^4 + (B - 1)x^3 + 1}$, is
- (A) $[3, \infty)$ (B) $[0, 3]$ (C) $[-3, 3]$ (D) $(-\infty, \infty)$

Paragraph for question nos. 21 to 23

Let $f : A \rightarrow B$ be an onto function defined as $f(x) = \frac{\sin^{-1} x + \tan^{-1} x}{\cos^{-1} x + \cot^{-1} x}$.

21. If the minimum and maximum value of $f(x)$ be m and M respectively then the value of $\frac{M}{m}$ is
- (A) -1 (B) -4 (C) -7 (D) Infinite

22. Let $g : B \rightarrow A$ be a function such that $g(f(x)) = x \quad \forall x \in A$ and $f(g(x)) = x \quad \forall x \in B$, then the value of $g(3)$ is
 (A) -1 (B) 0 (C) 0.5 (D) 1
23. The number of solutions of the equation $f(x^3 + 14x^2 + 13x - 5) = f(1 - x^2 + x^3)$ is
 (A) 0 (B) 1 (C) 2 (D) 3

Numerical based Questions :

24. If $\sum_{r=1}^{10} \tan^{-1}\left(\frac{3}{9r^2 + 3r - 1}\right) = \cot^{-1}\left(\frac{m}{n}\right)$ (where m and n are coprime), then find $(2m + n)$.
25. Perpendiculars are drawn from the angles A, B, C of an acute-angled triangle on opposite sides and produced to meet the circumscribing circle. If these produced parts are α, β, γ respectively, then the value of $\frac{(a/\alpha) + (b/\beta) + (c/\gamma)}{\tan A + \tan B + \tan C}$ is _____.

Matrix Match Type :

26. An function is defined on R or subset of R . Match the function given in column-I with their properties given in column-II. (Where $[x]$ denotes the greatest integer less than or equal to x .)

Column-I

- (A) $\sec^{-1}x$
 (B) $\sqrt{|x|}$
 (C) $\cos^{-1}(\cos x)$

Column-II

- (P) aperiodic
 (Q) into
 (R) odd or even
 (S) neither odd nor even

27. Match the following :

Column-I		Column-II	
A	The number of solutions to the equation $\frac{\pi}{2} \left(\left \sin^{-1}(\sin x) \right \right) = x^2 - \pi x $ are	p	4
B	The least value of the expression $2\log_{10} x - \log_x 0.01$ for $x > 1$ is	q	2
C	The number of common terms in the series 1, 4, 7, 10, 1391 and 4, 8, 16, 32 1024 are	r	5
D	Let A be a real number, and let $A = \frac{\sqrt{ x -2} + \sqrt{2- x }}{ 2-x }$ then the unit digit of $[A]^{2013}$ is (where $[.]$ denote greatest integer function)	s	3
		t	0

Code :

- (A) A→p; B→r; C→q; D→s
 (B) A→r; B→p; C→s; D→t
 (C) A→s; B→q; C→p; D→r
 (D) A→r; B→p; C→q; D→r

28. Match the following :

Column-I		Column-II	
A	If $p^2 - 2p \cos x = 673$ and $\tan \frac{x}{2} = 7$, the integral value of p is	p	8
B	If $\sin \theta + \cos \theta = m$ then the maximum value of m^2 is	q	9
C	r_1, r_2, r_3 are the radii of the circle drawn on the altitudes PD, PE and PF of $\triangle PBC, \triangle PCA, \triangle PAB$ respectively as diameter where P is the circumcentre of the acute angled $\triangle ABC$. The minimum value of $\frac{1}{18} \left[\frac{a^2}{r_1^2} + \frac{b^2}{r_2^2} + \frac{c^2}{r_3^2} \right]$ is (a, b, c are sides of $\triangle ABC$)	r	2
D	In $\triangle ABC$, $a = 6, b = 3$ and $\cos(A - B) = \frac{4}{5}$ then the area of the $\triangle ABC$ is	s	25

Code :

- (A) A→(s); B→(r); C→(p); D→(q)
 (B) A→(s); B→(p); C→(p); D→(q)
 (C) A→(r); B→(r); C→(p); D→(q)
 (D) A→(s); B→(q); C→(p); D→(r)

ANSWER KEY (CAPS - 4)

1. (D) 2. (D) 3. (D) 4. (B) 5. (D) 6. (A) 7. (C)
 8. (A) 9. (A) 10. (A) 11. (D) 12. (ABD) 13. (ABC) 14. (BCD)
 15. (ABD) 16. (ABD) 17. (BD) 18. (C) 19. (A) 20. (B) 21. (C)
 22. (D) 23. (B) 24. 32 25. (2.00) 26. (A) P, Q, S; (B) P, Q, R; (C) Q, R]
 27. (B) 28. (A)